Shayan Shakeri

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Toronto, Canada

PROFESSIONAL SUMMARY

AI Research Engineer working on reinforcement learning and hard-to-game evaluation: building VLM judges that turn model rollouts into dense, reward-gaming-resistant RL signal. Co-founder of Merlyn Labs; published RL/VLA research (8th place, Stanford BEHAVIOR-1K Challenge; LIBERO-PRO overfitting critique) and open-sourced RL-environment infrastructure; at BMO's AI Centre of Excellence, built a deterministic eval pipeline that surfaced systematic bias in a $200B+ AUM GenAI tool. Lifelong games enthusiast (D&D, Twilight Imperium) who approaches eval design as adversarial game construction.

CORE COMPETENCIES

Reinforcement Learning | Reward Modeling | RL Environments | Hard-to-Game Evaluation | VLM Judges | Large-Scale Experiments | Games + AI | PyTorch / JAX

WORK EXPERIENCE

Merlyn Labs
Aug 2025 - Present

AI Researcher & Co-Founder Toronto

- Developing **VLM judges** that score model rollouts into dense, context-dependent **RL rewards** designed to resist reward gaming.
- Open-sourced a flow-matching VLA integration for RLinf, enabling **RL training on the BEHAVIOR-1K environments** in OmniGibson (10,000+ demonstrations across 22 tasks).
- Placed **8th in Stanford's BEHAVIOR-1K Challenge**; identified proprioceptive collapse as a critical failure mode and published a technical report and LessWrong analysis.

BMO AI Centre of Excellence
Sep 2024 - Present

AI Research Engineer Toronto

- Built a **deterministic agent eval pipeline** from hundreds of synthesized inputs that uncovered systematic risk-downplaying bias in a GenAI tool serving **$200B+ AUM** wealth management.
- Building a graph-based agentic system answering multi-hop relational queries across bank client data (agent evaluation over reasoning paths).

Epineuron Technologies
May 2021 - Aug 2022

Biomedical & Electrical Engineering Co-op Mississauga

- Authored validation protocols (IEC 60601-1, ISO 13485) for PeriPulse, an FDA Breakthrough-designated device now in multi-national clinical trials.

PROJECTS

Stanford BEHAVIOR-1K Challenge
8th Place, Standard Track

Built and optimized RL environments and ran large-scale experiments (10,000+ demonstrations, 22 tasks in OmniGibson). Found chunked execution outperformed temporal ensembling 3x, with reward shaping tuned to resist gaming.

RL environments, Reward modeling, Large-scale experiments, OmniGibson

EDUCATION

M.Eng, MIE - AI & Robotics, University of Toronto — Sep 2024 - Apr 2027

Coursework: Deep Learning, Reinforcement Learning, AI Applications in Robotics, Healthcare Robotics.

B.Eng, Engineering Physics Co-op, McMaster University — Sep 2018 - Jun 2023

Dean's Honour List. Coursework: Computational Multiphysics, Quantum Computing, Numerical Methods, Signals & Systems.

SKILLS

ML & AI: Reinforcement Learning, Reward Modeling, RL Environments, LLM Fine-tuning, Hard-to-Game Evaluation, VLM Judges, Imitation Learning, Agents, Python, C/C++, PyTorch, JAX

Robotics & Simulation: VLA Models, OmniGibson, MuJoCo, Flow Matching, Sim2Real, ROS2